

Glossary for Genomics Experts

Exploring Deep Learning

1. Deep Learning (DL)

A subset of machine learning involving neural networks with multiple layers that can learn hierarchical representations from data.[ScienceDirect+1The Alan Turing Institute+1](#)

2. Neural Network

A computational model composed of interconnected nodes (neurons) organized in layers, inspired by the human brain's structure.[CSET+3Wikipedia+3The Alan Turing Institute+3](#)

3. Convolutional Neural Network (CNN)

A type of neural network particularly effective for spatial data, such as images or sequences, using convolutional layers to detect patterns.

4. Recurrent Neural Network (RNN)

Designed for sequential data, RNNs have connections that form directed cycles, allowing information to persist.

5. Transformer

An architecture that uses self-attention mechanisms to process sequential data, enabling parallelization and improved performance on tasks like language modeling.

6. Embedding

A representation of data (e.g., words, sequences) in a continuous vector space, capturing semantic relationships.

7. Fine-Tuning

The process of taking a pre-trained model and adjusting its parameters on a new, often smaller, dataset to specialize it for a specific task.

8. Retrieval-Augmented Generation (RAG)

A method that combines information retrieval with generative models, allowing the system to fetch relevant data and generate informed responses.

9. Explainable AI (XAI)

Techniques and methods that make the outputs of AI models understandable to humans, crucial in fields like healthcare for trust and transparency.[Wikipedia+1Wikipedia+1](#)

10. Overfitting

A modeling error where a model learns the training data too well, including its noise, leading to poor generalization to new data.